

Lecture 2 (13.10.2025)

Sequences

a sequence $\rightarrow$ an infinite set, indexed by natural numbers.

(x_n) $\rightarrow$ bounded $\begin{cases} \nearrow \text{above} \\ \searrow \text{below} \end{cases}$

$\rightarrow$ increasing $(x_{n+1} \geq x_n)$ / decreasing $(x_{n+1} \leq x_n)$

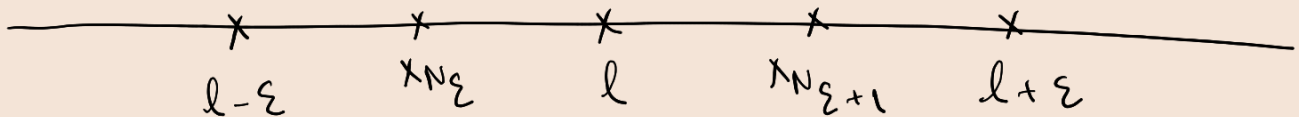
$\rightarrow$ monotone ($\nearrow$ or $\searrow$)

A sequence x_n has a limit if:

$(\forall) \forall \epsilon \in \mathbb{R}(\epsilon), \exists N_\epsilon \in \mathbb{N}$ s.t. $x_n \in V, (\forall) n \geq N_\epsilon$.

If $l \in \mathbb{R}$, x_n converges to l :

$(\forall) \epsilon > 0, \exists N_\epsilon \in \mathbb{N}$ s.t. $|x_n - l| < \epsilon, (\forall) n \geq N_\epsilon$.



x_n converges to $l \Leftrightarrow \lim_n |x_n - l| = 0$.

Ex. Prove that $\lim_n \frac{1}{n} = 0 = l$.

Let $\epsilon > 0$

We want $N_\epsilon \in \mathbb{N}$ s.t. $\frac{1}{n} - 0 < \epsilon, (\forall) n \geq N_\epsilon$.

$$\Leftrightarrow \frac{1}{n} < \epsilon$$

$$\Leftrightarrow \frac{1}{\epsilon} < n, (\forall) n \geq N_\epsilon.$$

Take $N_\varepsilon > \frac{1}{\varepsilon}$, then $n \geq N_\varepsilon > \frac{1}{\varepsilon}$, $(\forall) n \geq N_\varepsilon$

For example, $N_\varepsilon = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$.

- ① Every bounded seq. is convergent. F.
- * ② Every convergent seq. is bounded. T
- ③ Every monotone seq. is convergent. F
- ~ ④ Every convergent seq. is monotone. F
- ⑤ Every monotone seq. is bounded. F
- ⑥ Every bounded seq. is monotone. F.
- * ⑦ Every bounded + mon seq. is conv. T.

Prove that * ② Every convergent seq. is bounded.

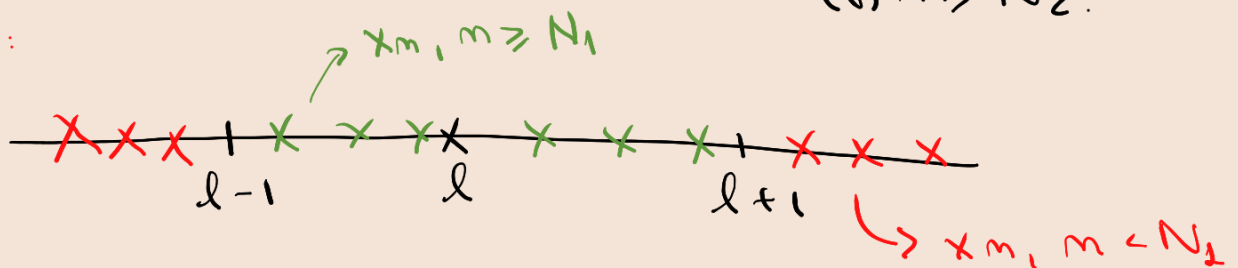
$(X_n) \text{ convergent} \Rightarrow (X_n) \text{ bounded}$

Def. of convergence: $(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N} \text{ s.t. } |X_n - l| < \varepsilon$

$\Leftrightarrow X_n \in (l - \varepsilon, l + \varepsilon),$

$(\forall) n \geq N_\varepsilon$.

$\varepsilon = 1$:



$$l = \lim_n X_n$$

Let $\varepsilon = 1$. Then $\exists N_1 \text{ s.t. } \underline{X_n \in (l-1, l+1), (\forall) n \geq N_1}$

BOUNDED

For $n \leq N_1$, X_n is also bounded by $\min, \max$.

⑦ Every bounded + mon seq. is conv. T.

Prove that * (7) Every bounded monotone sequence is convergent.

(x_n) bounded + monotone $\Rightarrow x_n$ convergent.

Say $(x_n) \nearrow$ and bounded.

$$\Rightarrow x_{n+1} \geq x_n, (\forall) n \in \mathbb{N}.$$

$$l = \sup x_n, n \in \mathbb{N} \text{ (exists because } x_n \text{ is bounded)}$$

! Any bounded set has a sup/inf.

Def. supremum: $(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N} : l - \varepsilon < x_n$

$$l - \varepsilon < x_{N_\varepsilon} \leq x_n$$

Then $l - x_n < \varepsilon, (\forall) n \geq N_\varepsilon.$

Weierstrass theorem:

monotone + bounded $\Rightarrow$ convergent.

Proposition:

any monotone sequence has a limit in $\overline{\mathbb{R}}$.

Sandwich theorem:

$$x_n \leq y_n \leq z_n, (\forall) n \geq n_0.$$

$$\text{and } \lim_n x_n = \lim_n z_n.$$

Proof

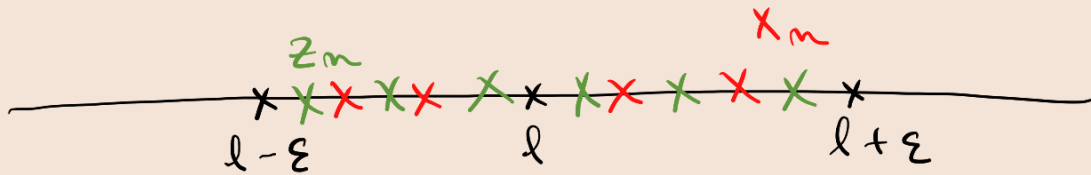
$$x_n \leq y_n \leq z_n, (\forall) n \geq n_0$$

$$l = \lim_n x_n$$

$$l = \lim_n y_n$$

$$y_n \rightarrow l: (\forall) \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } |x_n - l| < \varepsilon \quad (\forall) n \geq N$$

$x_n \rightarrow x$: $(\forall) \varepsilon > 0, \exists N_1 \in \mathbb{N}$ s.t. $|x_n - x| < \varepsilon, (\forall) n \geq N_1$.
 $z_n \rightarrow l$: $(\forall) \varepsilon > 0, \exists N_2 \in \mathbb{N}$ s.t. $|z_n - l| < \varepsilon, (\forall) n \geq N_2$.



$$x_n \leq y_n \leq z_n \quad | - l$$

$$x_{n-l} \leq y_{n-l} \leq z_{n-l}$$

$$\Rightarrow |y_{n-l}| \leq \max \{ |x_{n-l}|, |z_{n-l}| \}$$

$$\leq \varepsilon, \quad (\forall) n \geq \max \{ N_1, N_2 \}, n_0.$$

Cantor's nested intervals

$$a_n \nearrow, \quad b_n \searrow$$

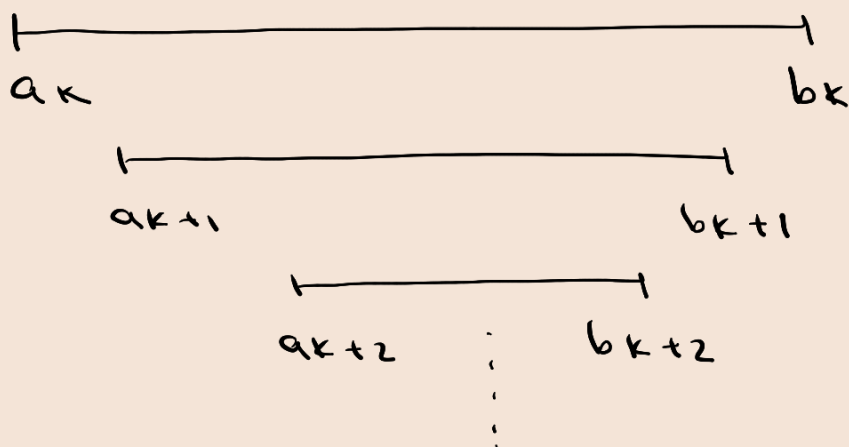
$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n, \quad (\forall) n \in \mathbb{N}$$

$$I_n = [a_n, b_n]$$

$$I_{n+1} \subseteq I_n.$$

$$\lim_n (b_n - a_n) = 0$$

$$\Rightarrow \exists x \in \mathbb{R} \text{ s.t. } \bigcap_{n=1}^{\infty} I_n = \{x\}$$



Proof:

Let $k \in \mathbb{N}$. Then $a_k \leq \sup A \leq b_k$ (1)

$a_k \in A$

b_k is an upper bound

Let $A = (a_m)$, $B = (b_m)$

$$a_k \leq \inf A \leq b_k \quad (2)$$

a_k is a lower bound

$b_k \in B$

(1): $a_k \leq \sup A \leq b_k$

$a_k \leq \inf B \leq b_k$

$\Rightarrow \sup A = \inf B = x$

$\Rightarrow \bigcap_{n=1}^{\infty} [a_n, b_n] = \{x\}$

$x = \lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} b_k$

$\lim_n x_n = l \in \overline{\mathbb{R}} \Leftrightarrow \lim_n \inf x_n = \lim_n \sup x_n = l.$

Ratio test

x_n s.t. $x_n > 0$

$\lim_n \frac{x_{n+1}}{x_n} = l \in \overline{\mathbb{R}}$

If $l < 1 \Rightarrow \lim_n x_n = 0$.

$l > 1 \Rightarrow \lim_n x_n = \infty$

$l = 1 \Rightarrow$ inconclusive.

Proof:

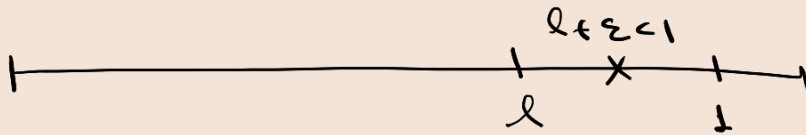
$\frac{x_{n+1}}{x_n} \rightarrow l < 1.$

By definition: $(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}$ s.t. $\frac{x_{n+1}}{x_n} - l < \varepsilon$

$$\Leftrightarrow \frac{x_{n+1}}{x_n} < l + \varepsilon$$

$$\Leftrightarrow \underline{x_{n+1} < (l + \varepsilon) \cdot x_n}$$

Let $\varepsilon > 0$ s.t. $l + \varepsilon \stackrel{\text{def.}}{=} q < 1$.



Then $x_{n+1} < q \cdot x_n$, $(\forall) n \geq N_\varepsilon$

$$x_n < q \cdot x_{n-1} < q^2 \cdot x_{n-2} < \dots < q^{n - N_\varepsilon} \cdot x_{N_\varepsilon}$$

$$\parallel$$

$$q^n \cdot \frac{x_{N_\varepsilon}}{q^{N_\varepsilon}}$$

$c = \text{cl.}$

$$\Rightarrow 0 < x_n < q^n \cdot c, (\forall) n \geq N_\varepsilon.$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & 0 & 0 \end{array} \quad (\text{because } q < 1) \quad \text{sub}^\infty \rightarrow 0.$$

Stolz-Cesàro

$(a_n), (b_n)$

1) $a_n \rightarrow 0$

$b_n \searrow 0$

or

2) $b_n \nearrow \infty$

$\frac{\infty}{\infty} \rightarrow \infty$

$\frac{0 \rightarrow 0}{0 \searrow 0}$

If $\lim_n \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = l$, then $\lim_n \frac{a_n}{b_n} = l$.

Bolzano-Weierstrass

Any bounded sequence has a convergent subsequence.

0.1. sequences

Cauchy sequence

(x_n) is called Cauchy if:

$$(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N} \text{ s.t. } |x_m - x_n| < \varepsilon, (\forall) m, n \geq N_\varepsilon.$$

any Cauchy sequence is bounded

Proof

Let $\varepsilon > 0$. Then $\exists N_1 \in \mathbb{N}$ s.t. $|x_m - x_n| < 1, (\forall) m, n \geq N_1$.
In particular, any $|x_m - x_{N_1}| < 1, (\forall) m \geq N_1$.

$\Rightarrow (x_n)$ is bounded, for $n \geq N_1$

$$x_n \in (x_{N_1} - 1, x_{N_1} + 1)$$

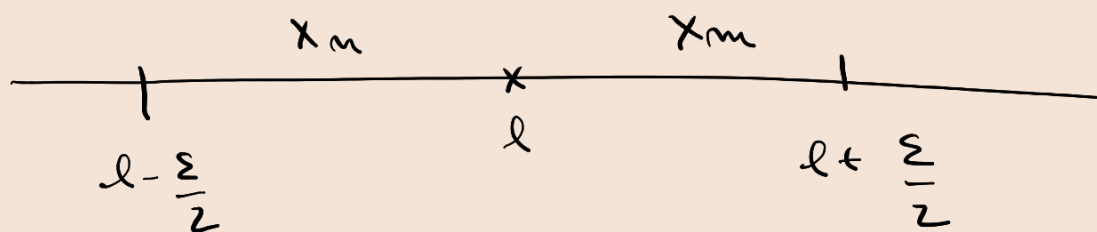
$\Rightarrow (x_n)$ is bounded, for $n < N_1$ (finite set)

a seq. is convergent $\Leftrightarrow$ that seq. is Cauchy

Proof: $\leftarrow$

$$(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N} \text{ s.t. } |x_n - l| < \frac{\varepsilon}{2}, (\forall) n \geq N_\varepsilon.$$

$$\begin{aligned} \text{I need: } |x_n - x_m| &< |x_n - l| + |l - x_m| \quad (\text{D.ing.}) \\ &< \varepsilon, (\forall) n, m \geq N_\varepsilon. \end{aligned}$$



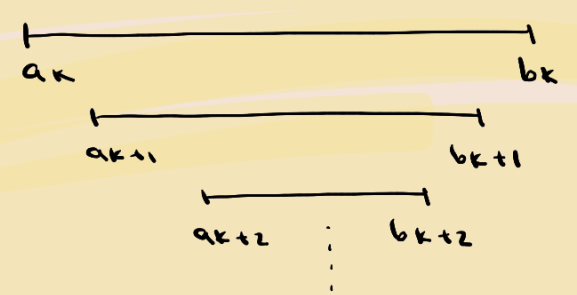
Summary:

1) (x_n) convergent $\Rightarrow (x_n)$ bounded

2) (x_n) bounded + monotone $\Rightarrow x_n$ convergent.

3) ! Any bounded set has a sup/inf.

4) any monotone sequence has a limit in $\overline{\mathbb{R}}$.

5) $\bigcap_{n=1}^{\infty} I_n = \{x\}$  Cantor's nested int.

6) $\lim_n x_n = l \in \overline{\mathbb{R}} \Leftrightarrow \lim_n \inf x_n = \lim_n \sup x_n = l$.

7) Ratio test ✓

8) Sandwich ✓

9) Any bounded sequence has a convergent subsequence. (B-W)

$$\frac{a_n}{b_n} \begin{cases} \rightarrow \frac{a_n \rightarrow 0}{b_n \rightarrow 0} \Rightarrow \frac{a_{n+1} - a_n}{b_{n+1} - b_n} \rightarrow \frac{a_n}{b_n} \text{ (Stolz-Cesaro)} \\ \rightarrow \frac{a_n}{b_n} \rightarrow \infty \end{cases}$$

10) $(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}$ s.t. $|x_m - x_n| < \varepsilon, m, n \in \mathbb{Z}$ (Cauchy seq.)

• any Cauchy sequence is bounded

a seq. is convergent $\Leftrightarrow$ that seq. is Cauchy

11) If $l \in \mathbb{R}$, x_n converges to l :

$(\forall) \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}$ s.t. $|x_n - l| < \varepsilon, (\forall) n \geq N_\varepsilon$.

12)

x_n converges to $l \Leftrightarrow \lim_n |x_n - l| = 0$.